

Head-Worn Displays for Emergency Medical Services

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ABSTRACT

In the prehospital environment, head-worn displays (HWDs) could support paramedics and emergency physicians during complex tasks and procedures. Previously, HWDs have been used in emergency medical service (EMS) contexts to support triage, telemedicine, patient monitoring, and patient localization. However, research on HWDs in EMS has three limitations: (1) HWD applications have not been developed based on field research of prehospital operations and training, (2) there are few guidelines that direct HWD deployment and application design, and (3) HWD applications seldom have been tested in randomized controlled trials. Therefore, it is unclear how HWDs affect EMS work and patient outcomes. During my PhD studies, I am investigating the potential of HWDs in EMS. I am addressing the limitations of previous research by conducting a literature review, a field study, design workshops, and a controlled evaluation study. The ultimate aims of this research are to benefit the work of EMS staff and to improve patient safety.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing; Interaction design.

KEYWORDS

Head-worn display, Augmented reality, Emergency medical services

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1 CONTEXT

I undertake my PhD studies at The University of Queensland, Brisbane, Australia under the supervision of Dr Ben Matthews and Professor Penelope Sanderson. I am in the mid-stage of my candidature which I started in April 2019 and which I am on track to complete in April 2022.

2 MOTIVATION AND PREVIOUS RESEARCH

Emergency medical services (EMS) have advanced clinical competencies and attend severe trauma cases in the field. However, the

prehospital environment imposes various challenges on EMS clinicians and paramedics, such as rapidly changing weather conditions, diverse medical needs of patients, and limited availability of clinical imaging and consultation. Although an increasing amount of research has been carried out to support EMS clinicians, the evidence that governs practice in the prehospital environment is still limited compared to evidence available for other medical disciplines [6]. Multiple studies have identified significant gaps in EMS research and they suggest research that should be undertaken in different areas to advance prehospital care [11, 21, 22]. The development and evaluation of decision support systems and patient information exchange systems have been classified as high priority research objectives [21].

However, accessing such systems via a smartphone or tablet computer might not always be possible in the busy EMS environment or might cause interruptions in patient treatment which can have negative consequences [23]. Head-worn displays (HWDs) might be a more suitable hardware platform for certain kinds of decision support and information systems [18] because they can provide the wearer with information in a continuous and hands-free fashion. In the following, when using the term ‘HWD’, I refer to monocular and binocular optical see-through HWDs. When using the term ‘HWD application’ I refer to non-spatial user interfaces that resemble two-dimensional smartphone applications and augmented reality (AR) user interfaces that map three-dimensional computer graphics onto the environment and adapt the view according to the user’s orientation.

Although most healthcare related HWD research has been conducted in hospitals [12], there is also a small corpus of work exploring the use of HWDs in the prehospital EMS environment. Primarily, HWD applications have been developed that connect a remote clinician with local EMS staff so to support the decision making of EMS staff in mass casualty situations (triage) [2, 5, 8], the preparation of hospital equipment prior to arrival of the patient [4], and the instruction of advanced patient interventions in the field [19]. Furthermore, HWDs can support the identification of potentially dangerous substances [2], the localization of a patient during search and rescue missions [16], the monitoring of vital signs [19], and the training of specific procedures [1].

3 GAPS IN PREVIOUS RESEARCH AND OBJECTIVES OF MY PHD PROGRAM OF RESEARCH

Although the preliminary research on HWDs in EMS looks promising, three gaps in the literature are apparent. First, it is unknown which tasks in EMS could be best supported by HWDs. In previous studies, use cases for HWDs in EMS have not emerged in a bottom-up way. Instead, applications have been developed in a top-down process by researchers who assumed that HWDs might

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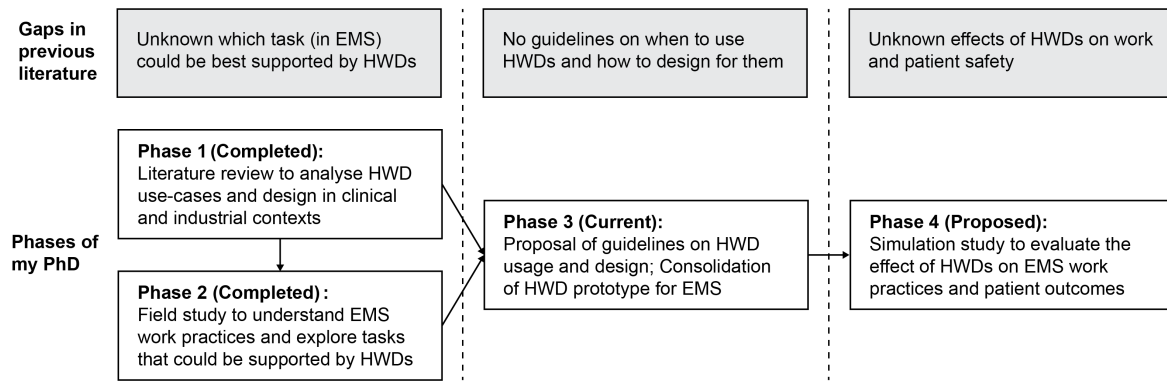


Figure 1: Illustration of my PhD program of research.

be beneficial in a certain context or situation. Second, there are no guidelines available that help clinical stakeholders decide when to deploy HWDs, and how to design them so that they support specific tasks. Third, rigorous evaluation of HWD applications in the EMS environment are rare. Because of this, it is unknown how HWDs affect EMS staffmembers' work in the prehospital environment and what the consequences for patient outcomes might be.

To address the aforementioned limitations in the literature, in my PhD research I will analyze which tasks can be supported well by HWDs in general and in EMS in particular (Phase 1 and 2; See Figure 1). Further, I will develop guidelines that can support the creation of usable HWD applications. By applying these guidelines, I will design a HWD based support system for EMS workers (Phase 3). Finally, I will evaluate the created HWD application in a high-fidelity simulation environment (Phase 4).

4 RESULTS TO DATE

In the following, I present the research that I have conducted over the past 18 months. I outline the main results of a literature review and an ethnographically informed workplace study (conditionally accepted for publication at CHI 2021 on 13 Dec 2020). Moreover, I detail how my work addresses gaps in previous literature.

4.1 Phase 1 – Literature Review: Head-Worn Display Applications and Design in Healthcare and Industry

In the first phase of my doctorate, I conducted a literature review. The purpose of the literature review was to uncover which tasks have been supported with HWDs in previous research. Furthermore, I explored how the design of HWD applications has been approached and what motivated their deployment. Because the two professional domains where the most HWD research has been conducted to date are healthcare and industry, I have focused on these contexts.

A scoping review [14] was carried out to locate relevant papers and generate insights about HWD use cases and design. In four databases (Web of Science, PubMed, IEEE Xplore, ACM Digital Library) I identified 351 papers that describe HWD applications in the healthcare or industrial context. A subset of 96 unique and highly cited papers was selected for a detailed analysis. During the

analysis, papers were clustered based on the properties of the HWD applications presented and specific information was elicited based on a custom coding scheme. The resulting review has two main contributions.

First, I identified seven distinct families of use cases for HWDs. They are (1) manual work guidance, (2) remote collaboration, (3) recording and preservation, (4) monitoring and notifying, (5) analysis, (6) understanding and visualization, and (7) controlling. I hope that the identification of the different use cases of HWDs will help researchers to locate previous publications that are relevant for their own HWD research—regardless of the context in which their work is situated. Second, I demonstrated that the design of HWDs has not been a primary research focus, and user centered design processes have seldom been applied to inform the development of HWD applications. To support the design of usable HWD applications in future research, I presented a comprehensive list of previous design research that has a high relevance for the development of HWD applications.

In the context of my PhD program of research, the results of the literature review directly affected the second and third phase of my project (see Figure 1). During the second phase of my PhD, when I studied work practices of EMS staff, the seven families of HWD use cases identified through the literature review helped me to generate a variety of HWD use cases that could support EMS operations. For example, knowing that manual work guidance HWD applications have been beneficial in other domains, I could envision a similar application for the EMS environment (see the following section for more details). In the third phase of my PhD, the summary of design research that can inform HWD design will help me to implement a usable HWD prototype. Furthermore, the design research reviewed could be the foundation for HWD specific design guidelines.

4.2 Phase 2 – Ethnographically Informed Workplace Study: HWD Use Cases to Support EMS Activities

In the second phase of my PhD, I studied EMS work practices in a field study. The purpose of the field study was to build on the previous literature review and identify concrete HWD applications that could support EMS staff during their work. To elicit real world needs of paramedics and emergency physicians I collaborated with

Table 1: Potential use cases for HWDs in EMS identified through the field study.

HWD use case	HWD description	Potential advantages	Potential disadvantages
Remote consultation	The HWD transmits first person video from the incident site and displays augmented reality markers controlled by a remote expert.	+ First person video could transmit a better impression of patient and incident site than oral descriptions + AR could boost the effectivity of instructing procedures	- Remote expert could be tempted to 'overcontrol' the incident site - Good network coverage required - Organizational structures required to facilitate HWD use
Clinical monitoring and imaging	The HWD displays the vital signs of the patient(s) at the incident site or the output of clinical imaging devices.	+ Vital signs could be monitored location independently + Information could be more legible than on conventional monitors that reflect sunlight + EMS staff could maintain an ergonomic posture	- In multiple casualty situations it might not be obvious whose data is displayed on the HWD - Discussing patient data with other staff might be difficult because only the HWD wearer can see the data
Training augmentation	The HWD instructs complex procedures that require a high level of skill but that are seldom performed. The HWD also increases the realism of low fidelity training scenarios by displaying distractor stimuli.	+ EMS staff who have previously worked in hospital environments could prepare for work in the complex prehospital environment + AR procedure guidance might be more accessible than written instructions	- Synchronizing distractor stimuli with simulation would be technologically challenging - Multiple HWDs had to display the same content because EMS staff often train in teams

the EMS provider Retrieval Services Queensland (RSQ). The collaboration allowed me to conduct situated interviews with EMS staff and observe training sessions. In total, I collected around 30 hours' worth of interview recordings and field notes. Through affinity diagramming [10] and thematic analysis [3] I generated the following insights.

To provide optimal care, EMS staff require as much patient data as possible. Ideally, the patient details are known long before arrival at the incident site. However, acquiring information can be challenging. Communication with local ambulance staff and organization structures can help to overcome information barriers. Moreover, preparation at the base, access to advanced retrieval equipment, and personal ingenuity are required to remain effective in the prehospital environment. Finally, experience is vital for efficient operations in the prehospital environment, but because the EMS staff that we accompanied only attended one or two missions per shift, they frequently participated in training sessions and exchanged information about previous missions with colleagues.

In the interviews, the participants also identified three use cases for HWDs that could be beneficial during their work. Table 1 provides a description of the use cases and advantages and disadvantages associated with them. Finally, I identified five factors that designers, developers, and clinical stakeholders have to consider before developing HWD applications for EMS. The factors are (1) the harsh outdoor environment in which HWDs will be used, (2) the extensive workload of staff, (3) the complex local and remote collaboration performed, (4) privacy requirements, and (5) the high variability of work.

The outcomes of the field study are particularly relevant for the third phase of my PhD program of research. The identified HWD use cases are the foundation for design activities that I will run with EMS staff. The design considerations that I described will also help me to develop a usable interactive HWD application.

5 CURRENT AND NEXT STEPS

Having presented the results of the first two phases of my doctorate, in the following subsections I outline my current research activities and next steps planned.

5.1 Phase 3 – Development of a HWD prototype and design guidelines in collaboration with EMS staff

Currently, in the third phase of my program of research, I am planning to engage in a participatory design process with the EMS staff members whom I accompanied during the field study. In particular, I will run five design workshops at the EMS base with a paramedic and an emergency physician participating in each session. The workshops will consist of three parts.

First, EMS staff will have the opportunity to experience and familiarize themselves with state of the art HWD hardware, such as the Microsoft HoloLens, Vuzix M300, and RealWear HMT1. Using the HWDs, participants will experience different games and productivity applications that feature, for example, video calling functionalities or heads-up note taking. I intend to use applications that feature different forms of user interaction (e.g. voice, gestures, hardware buttons) and different ways of visualizing information



Figure 2: Annotated screenshot of a first-person video of an EMS mission. The top right corner shows the potential user interface of a vital sign monitoring application. This photo was taken during the development of the workshop concept.

(e.g. AR or two-dimensional user interfaces). I am deliberately refraining from presenting medical HWD applications to not bias participants. After the familiarization with the HWD hardware, EMS staff will be able to rate which of the HWD use cases that were identified during the previous field study (see Table 1) has the highest potential to benefit their work. EMS staff will discuss the potential of HWDs during their work and write their thoughts about specific use cases on sticky notes that will be arranged as a small affinity diagram at the end of the session. In a next step, the participants and I will collaboratively design different aspects of the HWD user interface and generate paper prototypes. To do so, we will review first person videos of EMS missions. The required information during each phase of a mission and potential visualizations will be hand drawn on printed screenshots of the videos (see Figure 2). The videos will be sourced from the social media account of the EMS provider or from training sessions that are conducted at regular intervals.

The insights generated through the participatory design process will result in the development of a usable interactive HWD prototype that addresses the real-world needs of EMS staff. The HWD application will be evaluated in the fourth phase of my PhD. In addition, the design process will allow me to develop guidelines for other researchers who attempt to create HWD applications for EMS in particular and for other groups of users who operate in outdoor environments in general. The guidelines could include activities that proved to be useful to collaboratively design a HWD application or they could list different ways of visualizing information that participants liked and found effective.

5.2 Phase 4 – Evaluation of an interactive HWD application for EMS in a high-fidelity simulation

During the fourth phase of my program of research (commencing in Q2 2021) I will conduct a simulation study. In the simulation study, I will evaluate the developed HWD prototype without risk to

safety of patients or EMS staff members. The situation and task to be recreated in the simulator will be established during the development of the HWD application and depend on implemented functionalities and the purpose of the HWD.

To measure how the HWD might affect work in the prehospital environment, I will compare the HWD with a handheld tablet computer and with standard EMS practice (no support system) in a within design. I expect to collect quantitative measures such as task completion times, number of errors, situation awareness scores via SAGAT [7], or subjective workload scores via NASA TLX [9]. In addition to quantitative data, I will also collect qualitative data in semi-structured interviews. In these interviews, participants will be asked about the perceived usefulness of the HWD, but negative aspects of the HWD will be also explored, such as binocular rivalry [15], inattentive blindness [13], or HWDs' potential to distract [20] and to cause discomfort. The results of the simulation study will add to the corpus of existing HWD research in EMS and provide further evidence about the potential of HWDs to support work in the prehospital environment.

6 EXPECTED CONTRIBUTIONS OF MY PHD PROGRAM OF RESEARCH

My PhD program of research will make contributions to two fields: HCI (HWD) research and EMS research. My contribution to HCI research is significant because more and more businesses are interested in HWDs to optimize processes and to support workers in safety-critical and mission-critical environments. It has been estimated that smart glasses could save field service enterprises \$1 billion per year [17]. Clear evidence of when, where, and how HWDs confer benefits over other forms of information delivery will let decision makers more strategically deploy HWDs in healthcare but also in other work domains such as defence, construction, manufacturing, utilities, logistics, warehouse picking, transportation, and policing.

By placing my research in the EMS context, I will also contribute knowledge about the potential of HWD technology in this particular field. I hope that if HWDs could provide healthcare workers with relevant real-time data (and thus enable them to maintain situation awareness) and, for example, allow them to communicate with experienced physicians, patient safety could be significantly improved.

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